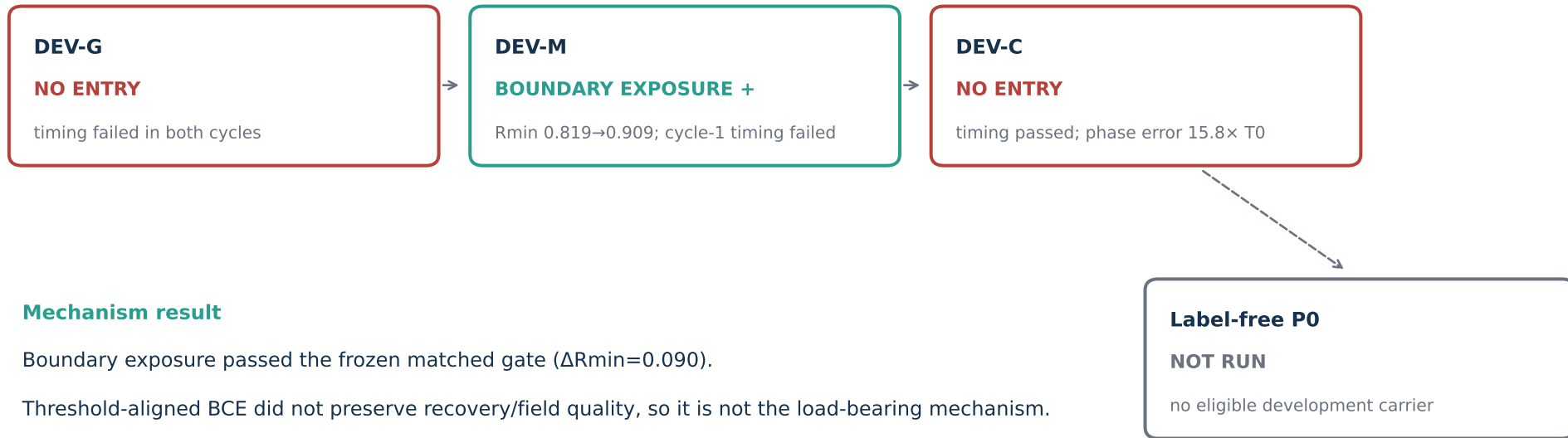


LF4 advances mechanism attribution but does not reach the physics-Pareto stage



Mechanism result

Boundary exposure passed the frozen matched gate ($\Delta R_{\min}=0.090$).

Threshold-aligned BCE did not preserve recovery/field quality, so it is not the load-bearing mechanism.

No selected carrier $\Rightarrow$ no physics-objective ratio, no PINN Pareto, and no candidate claim.